Executive Case Study Briefs: Multi-Echelon Logistics & Discrete Optimization Benchmarks

This dossier compiles verified official source links, organizational architectures, operational metrics, and mathematical/logistical constraints for two foundational supply chain case studies: **Barilla SpA (A)** and **Seven-Eleven Japan Co. (SEJ)**.

1. Barilla SpA (A)

Verified Official Sources & Catalog Links

- **Author:** Janice H. Hammond (Jesse Philips Professor of Manufacturing, Harvard Business School)
- **HBS Reference Number:** 9-694-046 (Originally Published: May 17, 1994; Revised: March 25, 2008; 21 pages)
- **Product Code:** `694046-PDF-ENG`
- **Official Primary Links:**
  - **Harvard Business Publishing (Education Catalog)** (Direct product page for academic and corporate course adoption)
  - **Harvard Business Review Store** (Direct individual purchase and PDF download)
  - **Harvard Business School Faculty Research Directory** (Official publication metadata and abstract)
  - **The Case Centre** (Product Reference: `694-046-1`)

Operational Context & Supply Chain Topology

- **Market Position (1990 Baseline):**
  - World's largest pasta manufacturer: **35% domestic Italian market share**, **22% total European market share**.
  - Brand portfolio: *Barilla* (32%), *Voiello* (Neapolitan high-end semolina, 1.5%), *Braibanti* (traditional Parma egg pasta, 1.5%), and *Mulino Bianco* (29% Italian packaged bakery market).

    - Geographic sales distribution: Approximately 50% in Northern Italy (higher penetration) and 50% in Southern Italy (larger per-capita pasta consumption, lower Barilla share).

- **Production Hub & Manufacturing Dynamics:**
    - **Pedrignano Plant (Parma):** World's largest pasta factory (1.25 million square meters site area), running 11 continuous automated lines (120 meters each), producing **900 metric tons (9,000 quintals) per day**.
    - **Manufacturing Constraints:**
        - Continuous sheet extrusion through bronze/Teflon dies, cutting, and hanging on dowels.
        - **4-hour tightly controlled drying cycle** with strict temperature and humidity curves inside tunnel kilns.
        - High changeover penalty: Switching pasta shapes (e.g., from thin spaghetti to hollow rigatoni) requires gradual thermal and humidity adjustments to prevent structural cracking. Lines run on rigid cyclic sequencing.
- **Network Nodes & Distribution Channels:**
    - **Two Central Distribution Centers (CDCs):**
        1. *Pedrignano CDC (Parma):* Handles Northern Italy distribution and all European exports.
        2. *Pozzilli CDC (Isernia):* Replenishes Central and Southern Italian distribution tiers.
    - **Italian Retail Fragmentation:** Italy operated over 100,000 independent small grocery grocers (*negozi tradizionali*), sharply contrasting with consolidated supermarket chains in the US or Northern Europe.
    - **Two Wholesale Tiers:**
        1. *Grande Distribuzione (GD) — Supermarket Chains (~35% of volume):* Large retail chains (e.g., Esselunga, Coop) operating centralized distribution hubs. Ordered full truckload (FTL) quantities directly from Barilla CDCs and cross-docked to their own stores.
        2. *Distribuzione Organizzata (DO) — Organized Wholesalers (~65% of volume):* Independent regional wholesalers and cooperative purchasing groups acting as intermediaries for tens of thousands of small neighborhood grocers who order small quantities several times a week.
- **SKU Diversity:**
    - **800 packaged dry pasta SKUs** (various shapes, thickness grades, branded carton boxes, cellophane bags, institutional foodservice packs).

- **200 fresh pasta SKUs** (refrigerated stuffed tortellini, ravioli; strict shelf-life decay and cold-chain handling).

Core Operational Problem Statement & Pain Points

- **Extreme Bullwhip Effect:**
  - Fork-end pasta consumption across Italian households was remarkably flat: **~18 kg per capita annually** (~350 grams/person/week) across all 52 weeks, with virtually zero seasonal fluctuation.
  - Despite steady consumer pull, weekly orders placed by distributors upon Barilla's CDCs swung wildly from **0 quintals to over 1,000–1,200 quintals per week** for single standard pasta SKUs.
  - Order variance upstream at the factory was 5x to 10x higher than end-consumer demand variance.
- **Simultaneous Stockouts & Excessive Inventory:**
  - Regional distributors carried an average of **2 to 3 weeks of finished goods inventory** (14 to 21+ days of forward supply).
  - Despite high safety stock buffers, distributors routinely experienced **10% to 13% stockout rates** on common dry pasta SKUs.
- **Driver 1 — Full Truckload (FTL) Discounts & Batching:**
  - Barilla offered price incentives and absorbed transit freight for full truckload orders (~30 pallets / 18,000 kg). Distributors held back orders until reaching FTL thresholds, causing intermittent demand surges.
- **Driver 2 — Canvass Promotional Cycles & Forward Buying:**
  - Barilla ran **10 to 12 promotional canvass periods per year** (each lasting 4 to 5 weeks).
  - During a canvass, sales reps offered **2% to 3% price discounts** plus performance volume bonuses.
  - Distributors engaged in speculative forward buying: placing massive multi-week orders during the final week of a canvass, followed by a 3-to-5-week post-promotional order dry-up where factory orders collapsed to near-zero while distributors burned through stockpiles.
- **Driver 3 — Long Lead Times:**
  - Lead times from distributor order placement to warehouse delivery ranged from **10 to 14 days**, inducing distributors to hoard safety stock whenever replenishment arrived late.

Core Mathematical & Operational Constraints

1. **Demand Volatility Amplification:** $\sigma^2_{\text{distributor}} \gg \sigma^2_{\text{consumer}}$ σdistributor2 ≫ σconsumer2, creating severe factory line setup inefficiencies and overtime labor costs.
2. **Batch Sizing Constraint:** Orders bounded by discrete pallet and FTL capacities ($\approx 18,000$ kg ≈ 18,000 kg increments).
3. **Data Silos & Boundary Resistance:** Distributors refused to share Point-of-Sale (POS) or warehouse stock levels, viewing inventory ownership and order timing as their core commercial prerogative.
4. **Lead Time Buffer:** Replenishment cycle of $L = 10$ to 14 daysL = 10 to 14 days requiring safety stocks of $SS = z \cdot \sigma_L \cdot \sqrt{L}$SS = z · σL · L

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2. Seven-Eleven Japan Co. (SEJ)

Verified Official Sources & Catalog Links

- **Author:** Sunil Chopra (IBM Distinguished Professor of Operations Management & Information Systems, Kellogg School of Management, Northwestern University)
- **Kellogg Reference Number:** Case No. `KEL026` (Originally Published: 2003 / 2005; Revised: May 27, 2022; 14 pages)
- **Product Code:** `KEL026-PDF-ENG`
- **Official Primary Links:**
  - **Harvard Business Publishing (Education Catalog)** (Direct product page in HBSP collection)
  - **Harvard Business Review Store** (Direct individual purchase and PDF download)
  - **Kellogg Case Collection via Emerald Publishing** (DOI-authenticated case record)
  - **The Case Centre** (Product Reference: `606-037-1`)

Operational Context & Supply Chain Topology

- **Network Scale & Dominance:**
  - Expanded from 9,000 retail locations to **over 15,000 stores across Japan** (>21,000 today).
  - Over 95% of retail stores operate under franchise agreements.

    - **Area Dominance Strategy:** SEJ enters a target geographic micro-region and rapidly clusters **50 to 60 stores within a tight radius**. This maximizes consumer visual presence, shortens inter-store transit times on delivery loops, and concentrates local distribution center efficiency.

- **Store Physical Constraints:**
    - Typical store footprint is **only 100 to 130 square meters** (~1,080 to 1,400 sq ft).
    - Carries an active assortment of **2,500 to 3,000 distinct SKUs**.
    - Backroom space is strictly minimized to **less than 10%–15% of floor area**, housing only employee lockers and the Store Computer terminal.
    - **Zero Backroom Inventory Floor:** Stores hold zero reserve warehouse stock. Every carton received from a delivery truck must be placed directly onto front display shelves or reach-in cooler racks. If an item cannot fit on the shelf, it cannot be received.
- **Manufacturing & Cross-Dock Hubs:**
    - SEJ owns virtually no manufacturing facilities or delivery vehicles on its corporate balance sheet.
    - **Nihon Delica Foods Association (NDF):** A network of over 170 dedicated third-party manufacturing plants producing fresh food exclusively for SEJ.
    - **Combined Distribution Centers (CDCs):** Third-party logistics hubs that consolidate supplier loads. Suppliers make bulk drops into the CDC; the CDC cross-docks, sorts by delivery route, and dispatches multi-temperature consolidated trucks.
    - Daily truck visits per store were reduced from **~70 trucks/day in the 1970s to fewer than 10–12 trucks/day** under the Combined Delivery System.

Temperature Regimes & Multi-Wave Delivery Cadence

SEJ partitions its entire product portfolio into four temperature zones, each operating on dedicated logistics channels:

Temperature Zone	Target Range	Product Categories	Delivery Cadence
Frozen	−20°C −20∘C	Ice cream, frozen novelties, packaged ice, frozen meals	3 to 7 drops / week (seasonal pacing)
Chilled	+5°C +5∘C	Sandwiches, rice balls (*onigiri*), bento lunchboxes, salads, chilled noodles, fresh dairy	**3 to 4 drops daily**
Ambient	+20°C +20∘C	Canned beverages, instant ramen, dry confectionery, baked goods, magazines, sundries	1 drop / day
Warm / Heated	+55°C +55∘C	Canned hot tea/coffee, hot counter food (*nikuman* buns, *oden*, fried chicken)	Delivered hot or held in display heating units

The 3-Wave Daily Delivery Windows for Chilled Fresh Food:

- **Wave 1 (Breakfast / Lunch Peak):**
  - Prepared in dedicated commissary kitchens between 3:00 AM and 6:00 AM.
  - Arrives at stores **between 10:00 AM and 11:00 AM** to fully stock shelves prior to the 11:30 AM – 1:30 PM lunch rush.
- **Wave 2 (Afternoon Snack Peak):**
  - Arrives at stores **between 3:00 PM and 4:00 PM** (fresh desserts, pastries, sandwiches).
- **Wave 3 (Dinner / Evening Commute Peak):**
  - Cooked in early afternoon.
  - Arrives at stores **between 7:00 PM and 8:00 PM** (evening bento boxes, dinner side items, prepared hot food).
- **Late-Night Cadence (Ambient & Magazines):**
  - Arrives **between 11:00 PM and 2:00 AM** during minimum road traffic.
- **Delivery Window Tolerance:** Delivery trucks are allocated tight **10-to-15-minute arrival windows**. To avoid noise and air pollution in residential neighborhoods, drivers turn off vehicle engines and scan crate barcodes within 10 minutes.

Information Architecture & Ordering Mechanics

- **Total Information System (TIS):**

    - **POS Cash Register:** Cashiers scan item barcodes and simultaneously log customer demographic tags:
        - *Gender:* Male / Female.
        - *Age Bracket:* Child (<13), Teen (13–19), Young Adult (20–39), Middle-Aged (40–59), Senior (60+).
        - Data captured: Exact timestamp of purchase, weather conditions, SKU combinations.
    - **Graphic Ordering Terminal (GOT):** Handheld terminal carried by store clerks in the aisles. Displays current on-shelf inventory, hourly sales history, past 10 weeks of ordering records, local micro-weather forecasts, and historical discard rates per SKU.
    - **Store Computer (SC):** Local node aggregating store telemetry and sending batch replenishment requests directly to CDCs and factory production schedulers via high-speed network.
- **Item-by-Item Ordering (*Tanpin Kanri* — 単品管理):**
    - Store ordering is not algorithmic or centrally dictated; it is deliberately delegated to individual frontline clerks who manage specific shelf categories.
    - **Hypothesis Formation (*Kasetsu*):** Clerk reviews tomorrow's temperature forecast, precipitation probabilities, and local neighborhood events (school sporting meets, construction projects). The clerk forms a hypothesis: "*Rain and a 5 deg C temperature drop tomorrow will reduce cold noodle sales by 40% and increase hot noodle soup and oden by 60%.*"
    - **Verification (*Kensho*):** The next day, POS logs are evaluated against the hypothesis: Did hot items sell out early (opportunity loss)? How many unsold items expired and were discarded (waste loss)? The clerk refines the ordering pattern for the next cycle.

Spoilage Clocks, Discard Discipline & Opportunity Cost Philosophy

- **Hour-Specific Expiration:** Fresh items carry printed expiry timestamps down to the exact hour and minute (16-to-24-hour stability window).
- **Automated POS Lockout:** If a customer brings an expired food item to the checkout counter, the barcode scanner generates an audible error and **locks the transaction**. The system physically refuses to sell expired food.
- **Shelf Sweep Cadence:** Store staff conduct manual shelf sweeps **1 to 2 hours before the printed expiration timestamp**, removing expiring items before customers encounter them.
- **Stockout vs. Discard Priority:**

- Chairman Toshifumi Suzuki's operational doctrine: **A stockout (lost opportunity) is more damaging to enterprise value than a product discard.** A stockout drives the customer to a competing chain (Lawson, FamilyMart).
- SEJ enforces a disciplined discard target of **1.5% to 2.5% of fresh food gross sales**, treating discard expense as an investment in 100% on-shelf availability and guaranteed freshness.

3. Comparative Operational Matrix

Operational Dimension	Barilla SpA (A) (HBS 9-694-046)	Seven-Eleven Japan Co. (Kellogg KEL026)
Core Supply Chain Paradigm	Batch-Push / Disconnected Wholesale	Ultra-Responsive Pull / Direct-to-Shelf Flow
Manufacturing Hubs	Flagship Pedrignano plant: 1.25M m2, 11 lines, 900 tons/day	170+ dedicated partner factories (NDF consortium)
Distribution Nodes	2 Central Distribution Centers (Pedrignano & Pozzilli)	200+ Cross-Dock Combined Distribution Centers (CDCs)
Delivery Frequency	Weekly / bi-weekly batch orders; 10–14 day lead times	**3 to 4 daily delivery waves** aligned with meal peaks
Inventory Buffer Policy	Distributors hold **2 to 3 weeks of stock**; 10%–13% stockouts	**Zero backroom storage**; direct-to-shelf allocation
Order Generation	Distributor batching driven by FTL discounts & canvass deals	Clerk item-by-item ordering (*Tanpin Kanri*) via handheld GOTs
Demand Distortion	Severe Bullwhip effect driven by trade promotions & forward buying	Real-time POS telemetry paired with customer demographics
Shelf-Life Horizon	Ambient dry pasta (months to years shelf life)	Fresh prepared foods (16 to 24-hour expiration clocks)
Core Mathematical Tension	FTL batch economies vs. setup costs & inventory carrying cost	Hourly transport routing under vehicle time windows & discard limits

Document compiled strictly detailing empirical problem frameworks, operational parameters, and physical constraints. Formatted for print or PDF conversion.